

Fleet Observability & Incident Investigation

Executive Summary Report — Data Analytics & Observability Internship

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INCIDENT
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WINDOW
10:00 AM – 11:00 AM

1 Project Objective

To conduct a rigorous, data-driven postmortem on a critical operational incident affecting a fleet of Autonomous Mobile Robots (AMRs). The goal was to move beyond assumptions and use **raw telemetry, trip lifecycle records, and system event logs** to precisely reconstruct the failure sequence, identify all fleet-wide anomalies, and conclusively prove the root cause of the mission-critical trip cancellation.

2 Approach

Phase	Module	Description
1 Data Profiling	preprocessing.py	Assessed dataset health — checked missing values, timestamp integrity, and row uniqueness across all three data sources before any analysis.
2 Exploratory Analysis	exploratory_analysis.py	Visualized CPU, battery, and network latency distributions fleet-wide to isolate statistical outliers.
3 Timeline Reconstruction	incident_timeline.py	Fused three asynchronous data streams (telemetry, trips, events) into a single, second-level chronological timeline.
4 Root Cause Extraction	root_cause_analysis.py	Applied rule-based pattern matching on the fused timeline to link prior events to trip cancellation outcomes.

3 Incident Timeline (10:00 AM – 11:00 AM)

Time	Robot	Event	Details
10:05:12	Robot_01	Trip T101 Started	Source to Destination
10:05:14	Robot_01	Resource Lock Acquired	Zone_A secured
10:05:55	Robot_01	Battery Low Warning	Pre-trip check missed
10:06:10	Robot_01	NETWORK LATENCY SPIKE	1450 ms sustained
10:06:18	Robot_01	RESOURCE LOCK TIMEOUT	Zone_B -- failed to acquire
10:06:20	Robot_01	TRIP T101 CANCELLED	Safety timeout triggered
10:15:00	Robot_02	Trip T102 Started	Duplicate log entry observed
10:17:40	Robot_03	Trip T204 Started	Zone_C lock acquired
10:18:05	Robot_03	Position Jump Detected	Coords: X:999, Y:999
10:18:15	Robot_03	Trip T204 Completed	Self-recovered successfully

Time	Robot	Event	Details
10:21:30	Robot_02	CPU Saturation	95% CPU load
10:24:00	Robot_02	Trip T102 Completed	Resilient under compute load
10:40:00	Robot_04	Trip T301 Started	Normal start
10:44:00	Robot_04	Battery Level Jump	32% to 41% (impossible mid-trip)

Critical Cascade: The 1450ms network spike at 10:06:10 directly caused the Zone_B lock timeout 8 seconds later, which triggered the automatic safety mechanism cancelling Trip T101. The **entire failure chain spanned only 10 seconds**.

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Five Detected Anomalies

ANOMALY 1 |
CRITICAL

Network Latency Spike — Robot_01

Observed: Network latency of **1450 ms** at 10:06:10, confirmed in both event logs and telemetry.
Threshold: Normal operations require <200 ms for lock acquisition RPC calls.
Impact: Mission-critical. Directly caused Trip T101 cancellation.

ANOMALY 2 |
WARNING

CPU Saturation — Robot_02

Observed: CPU utilization reached **95%** at 10:21:30, above the 90% alert threshold.
Impact: Non-critical. Robot_02 successfully completed Trip T102, demonstrating edge-software resilience under extreme compute pressure.

ANOMALY 3 |
WARNING

SLAM Localization Glitch — Robot_03

Observed: Telemetry logged coordinates **X:999, Y:999** at 10:18:05 — a sentinel value indicating an unhandled null or out-of-bounds SLAM output.
Impact: Non-critical. Robot self-recovered within 10 seconds and completed Trip T204.

ANOMALY 4 |
HARDWARE

Impossible Battery Jump — Robot_04

Observed: Battery level rose from **32% to 41%** during an active running trip at 10:44:00, confirmed by a BatteryLevelJump event log entry.
Impact: Hardware-critical. AMRs cannot charge while in motion. This isolates a defective Battery Management System (BMS) sensor requiring replacement.

ANOMALY 5 |
SOFTWARE

Event Log Idempotency Bug — Robot_02

Observed: The TripStarted event for Trip T102 was logged **twice** at exactly 10:15:00.
Impact: Non-critical operationally, but this indicates a bug in the telemetry ingestion pipeline that could corrupt downstream analytics, aggregations, and billing counts.

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Root Cause

Root Cause: Network-Induced Resource Lock Timeout

A severe network latency spike (1450ms) on Robot_01 prevented it from communicating with the resource management server within the required timeout window. The server, unable to receive lock acquisition confirmation for Zone_B, triggered an automatic safety timeout and cancelled Trip T101.

Compute was stable at 54% CPU. Battery level was 76%. The failure was purely a network-layer issue.

Confirmed Evidence Chain:

#	Timestamp	Event	Proof
1	10:06:10	NetworkLatencyHigh (1450ms)	Event log + telemetry data both confirm
2	10:06:18	ResourceLockTimeout (Zone_B)	8 seconds after latency spike
3	10:06:20	TripCancelled (T101)	2 seconds after lock timeout

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Additional Observation — Hidden Safety Risk

Safety Design Flaw Identified

At 10:05:55, Robot_01 generated a **BatteryLowWarning** and yet the system allowed the robot to continue its trip, acquire Zone_A, and attempt to acquire Zone_B.

Why this is dangerous: In a warehouse environment, if a low-battery robot acquires a lock on a shared critical zone and its battery dies mid-transit, it creates a **physical blockade**. No other robot can enter that zone until the lock is manually released and the robot is physically removed — potentially **shutting down the entire fleet's throughput**.

Recommendation: Implement a pre-trip battery gate — any robot below a configurable threshold (e.g., 30%) should be barred from acquiring new zone locks and immediately routed to a charging dock.

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Conclusion

This investigation conclusively identified a **network-induced resource lock timeout** as the sole cause of the mission-critical Trip T101 cancellation. The four secondary anomalies were isolated to individual robots and did not affect fleet-wide operations — a testament to the robustness of the edge-deployed robot software.

Finding	Severity	Status	Recommended Action
Robot_01 Trip Cancellation	CRITICAL	Root cause proven	Network SLA audit + redundancy review
Robot_02 CPU Saturation	WARNING	Edge SW resilient	Set capacity alerts; monitor trends
Robot_03 SLAM Glitch	WARNING	Self-recovered	Improve SLAM null/boundary handling
Robot_04 BMS Sensor Fault	HARDWARE	Defect confirmed	BMS sensor replacement & calibration
Pipeline Idempotency Bug	SOFTWARE	Defect confirmed	Fix ingestion deduplication logic
Low-battery lock policy	SAFETY	Gap identified	Implement pre-trip battery gate ASAP

Key Takeaway: The most urgent systemic risk is not the network incident itself, but the **absence of a battery-level pre-check** before zone lock acquisition — a policy gap that could cause a significantly more severe operational disruption and full fleet gridlock if left unaddressed.